

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis With SAS/SPSS/R
Practical No 14

Aim :- Performing logistic regression using glm() (R)

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Project: (None)
Source
Console Terminal Background Jobs
R 4.52.0 ~ /
> # Read the dataset
> salary_data <- read.csv("ds_salaries.csv")
>
> # Take top 5 rows ONLY
> salary_data <- head(salary_data, 5)
>
> # View data
> head(salary_data)
  x work_year experience_level employment_type job_title salary salary_currency salary_in_usd employee_residence
1 0 2020 MI FT Data Scientist 70000 EUR 79833 DE
2 1 2020 SE FT Machine Learning Scientist 260000 USD 260000 JP
3 2 2020 SE FT Big Data Engineer 85000 GBP 109024 GB
4 3 2020 MI FT Product Data Analyst 20000 USD 20000 HN
5 4 2020 SE FT Machine Learning Engineer 150000 USD 150000 US
  remote_ratio company_location company_size
1 0 DE L
2 0 JP S
3 50 GB M
4 0 HN S
5 50 US L
> str(salary_data)
'data.frame': 5 obs. of 12 variables:
 $ x : int 0 1 2 3 4
 $ work_year : int 2020 2020 2020 2020 2020
 $ experience_level : chr "MI" "SE" "SE" "MI" ...
 $ employment_type : chr "FT" "FT" "FT" "FT" ...
 $ job_title : chr "Data Scientist" "Machine Learning Scientist" "Big Data Engineer" "Product Data Analyst" ...
 $ salary : int 70000 260000 85000 20000 150000
 $ salary_currency : chr "EUR" "USD" "GBP" "USD" ...
 $ salary_in_usd : int 79833 260000 109024 20000 150000
 $ employee_residence : chr "DE" "JP" "GB" "HN" ...
 $ remote_ratio : int 0 0 50 0 50
 $ company_location : chr "DE" "JP" "GB" "HN" ...
 $ company_size : chr "L" "S" "M" "S" ...
>
> # Create binary target variable
> salary_data$Approved <- ifelse(
+ salary_data$salary_in_usd > median(salary_data$salary_in_usd),
+ 1,
+ 0
+ )
>
> salary_data$Approved <- as.factor(salary_data$Approved)
>
> # Logistic Regression Model
> log_model <- glm(
+ Approved ~ remote_ratio + experience_level + employment_type,
+ data = salary_data,
+ family = binomial
+ )
Error in contrasts<-"("tmp", value = contr.funs[1 + isOF[nn]]) :
contrasts can be applied only to factors with 2 or more levels
```

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